

Mapping the UK Climate & Nature Policy Landscape for a new Met Office UK marine heatwave product

Potential UK climate – nature policy applications · Gap assessment · Prepared July 2026

Key points

- **This product does not yet exist as an operational Met Office service.** This is a gap assessment: the external literature (Holbrook 2019; Oliver 2018; Cornes/MCCIP 2025) shows marine heatwaves (MHWs) are now well characterised and their ecological and fisheries impacts are established, and this asks what UK policy could do with a national, policy-facing MO MHW product.
- **The building blocks already exist at the Met Office.** OSTIA (daily satellite+in-situ sea surface temperature), the FOAM / Copernicus Marine North-West Shelf system (3D temperature, incl. sub-surface and seafloor) and the ESA SST_cci record are all MO-run. The gap is not raw data but an **operational, categorised, forecast MHW product** (event onset, duration, category I–IV, sector impact layers) — none exists for UK policy today.
- **Highest-value prospective applications (Relevance 3):** the Marine Climate Change Impacts Partnership (MCCIP) sea-temperature evidence stream, the UK Marine Strategy 2025 Programme of Measures (Good Environmental Status), the Fisheries Act 2020 climate-change objective / Fisheries Management Plans (†), and CCRA4 (Met Office-led Technical Report, mid-2026; CCRA4 laid Jan 2027).
- **Timely:** Cefas and the Met Office are already building UK MHW-prediction capability — the *40 priority questions* horizon-scan (Pinnegar, Rowland et al., npj Ocean Sustainability 2025) and the Defra Marine Science programme (Sep 2024) — the natural home for an operational product.
- **Maturity varies:** MHW monitoring itself is **Demonstrated** (EEA/Copernicus/MCCIP already publish UK & European SST and MHW indices); operational forecasting into fisheries, aquaculture and adaptation processes is **Emerging**; marine-planning, protected-area and blue-carbon uses are **Conceptual**.
- **Caveats:** the bibliography is an external MHW reading list, not an MO publication list, so MO capability was scoped from MO/Cefas/Copernicus sources; and three of the highest-value marine activities — the Fisheries Act/JFS/FMPs, aquaculture food-safety (HAB/*Vibrio*) monitoring, and the UK Blue Carbon Evidence Partnership — are operational processes not held in the governance workbook (marked †).

The evidence base for a prospective UK marine heatwave product

A UK marine heatwave product would deliver operational, categorised ocean-heat information — daily sea surface temperature (SST) anomalies and MHW category (Hobday I–IV: moderate to extreme), sub-surface and seafloor temperature / ocean heat content, and short-range to seasonal forecasts of MHW onset, duration and decline — across the UK Exclusive Economic Zone and North-West European shelf at a few-kilometre resolution, on a 1982-to-present satellite baseline. The Met Office already runs the underpinning systems (OSTIA SST; the FOAM / Copernicus Marine North-West Shelf model; the ESA SST_cci climate record) and co-leads UK MHW science with Cefas. The external evidence base below shows the policy-relevant applications are established science elsewhere.

- **Intensifying UK MHWs** — around UK and Irish seas MHWs have grown in frequency, duration and magnitude; projections indicate coastal regions could be in MHW conditions for more than half the year by 2100 under high emissions (Cornes et al. / MCCIP 2025).
- **Fisheries and distribution shifts** — MHWs drive shifts in the distribution and productivity of commercially important fish and shellfish, directly relevant to stock assessment and fisheries management (Holbrook et al. 2019; Plecha et al. 2021).
- **Habitat damage** — MHWs cause measurable stress and mortality in seagrass, kelp and other temperature-sensitive habitats, with knock-on effects on blue-carbon storage (Jung et al. 2023; Ford et al. 2025).

- **Compound and sub-surface events** — coastal MHWs can be generated or amplified by atmospheric heat and stratification, and seafloor/sub-surface heatwaves with non-summer impacts are poorly captured by SST-only monitoring (Dzwonkowski et al. 2020; European Commission 2026).

A prospective MO UK marine heatwave product would translate these demonstrated capabilities into a national, policy-ready evidence and early-warning layer; the table maps where that would matter and how proven each use is.

Policy applications, relevance and maturity

Two independent axes. **Relevance** (materiality of a UK marine heatwave product to the policy): **3 High** = would be a direct, central input; **2 Medium** = a useful supporting evidence layer; **1 Low** = marginal or indirect. **Maturity** (how proven the application is): Demonstrated = proven elsewhere or already used and transferable; Emerging = partial evidence, needs UK development/validation; Conceptual = plausible but unproven for this use. Items marked † are activities not held in the governance workbook. Policy names link to the source page.

Ref	Policy activity	Relevance	Maturity	What a UK marine heatwave product could enable for this policy
POL-049	Marine Climate Change Impacts Partnership (MCCIP)	3 HIGH	Demonstrated	The established research→policy channel for UK marine climate evidence. MCCIP already publishes a Sea Temperature evidence stream (rolling updates; Annual Summary 2025) that MO ocean data feed, and which underpins Defra/agency and UK Marine Strategy assessments. A dedicated MHW product would slot directly into this route — the clearest and most proven application.
POL-028	UK Marine Strategy 2025 — Programme of Measures	3 HIGH	Emerging	The Part One 2025 assessment explicitly flags ocean warming and marine heatwaves as risks to achieving Good Environmental Status. An MHW product would provide the temperature/ocean-heat evidence layer for the pelagic-habitat, food-web and fish-stock GES descriptors — evidenced in the strategy text but not yet a formal operational input.
†	Fisheries Act 2020 climate-change objective / Joint Fisheries Statement & Fisheries Management Plans	3 HIGH	Emerging	The Act's climate-change objective and the FMPs must account for climate-driven shifts in stock distribution and productivity. MHW monitoring and forecasts could inform stock advice, seasonal management and the ecosystem-based approach set out in the 2022 Joint Fisheries Statement. Not held as a workbook record (†); MCCIP already issues fish/fisheries/aquaculture policy briefs on this.
POL-017	Climate Change Risk Assessment (CCRA3 2022; CCRA4-IA 2026, CCRA4 laid Jan 2027)	3 HIGH	Emerging	Marine and fisheries climate risks are central CCRA risks and the Met Office leads the CCRA4-IA Technical Report (publishing mid-2026). An operational MHW product is a high-value, near-term input to quantify ocean-heat exposure for CCRA4 and the subsequent NAP — though present assessments rely on SST trends and projections rather than a categorised MHW product.
POL-002	National Adaptation Programme (NAP3) 2023–2028	2 MED	Emerging	Translating CCRA marine risks into action would benefit from spatially explicit ocean-heat and MHW information to prioritise fisheries, aquaculture and habitat-resilience measures; the evidence exists but is not yet a defined NAP input.
POL-001	Environmental Improvement Plan 2025 (marine environment goals)	2 MED	Emerging	The EIP's clean-and-healthy-seas and marine-protected-site goals, and the associated EIF marine-condition indicators, are sensitive to ocean warming and MHWs. An MHW product could attribute temperature-driven condition change in the pelagic-habitat and fish/shellfish indicators; not yet an input.
POL-003	UK National Biodiversity Strategy and Action Plan 2025	2 MED	Emerging	Delivers GBF targets on climate-change pressure (Target 8) and invasive non-native species (Target 6). MHW and ocean-heat information could help anticipate thermally driven range shifts and which marine INNS become establishment-capable under UK warming.
†	Aquaculture & shellfish food-safety monitoring (Cefas – FSA / Food Standards Scotland; HABs and Vibrio)	2 MED	Emerging	MHWs are linked to harmful algal blooms and elevated pathogenic <i>Vibrio</i> in UK shellfish waters (Cefas, 2018 event). MHW forecasts could pre-warn biotoxin and <i>Vibrio</i> risk for the biotoxin-monitoring and classification programmes protecting shellfisheries and public health. Operational process not held in the workbook (†).

Ref	Policy activity	Relevance	Maturity	What a UK marine heatwave product could enable for this policy
POL-025	UK National Climate Science Partnership (UKNCSP)	2 MED	Conceptual	Met Office-led coordinating mechanism for translating climate science into CCRA/NAP and policy. It is a natural, but as yet latent, route to channel operational ocean-heat/MHW outputs into marine policy — an enabler rather than a defined application.
POL-026	Natural Capital & Ecosystem Assessment (NCEA) Programme	2 MED	Conceptual	NCEA builds national marine and coastal natural-capital baselines of extent and condition. Ocean-heat and MHW exposure could feed a thermal-stress condition layer for marine habitats, but its use in condition accounts is not yet defined.
POL-059	UK Marine Policy Statement and Marine Plans	2 MED	Conceptual	Marine spatial planning decisions (aquaculture siting, MPA management, developments) could screen thermal exposure and MHW risk; plausible but not an articulated function of the plans.
POL-060	Highly Protected Marine Areas (HPMAs)	2 MED	Conceptual	The three designated HPMAs (2023) and future sites aim to build ecosystem resilience, partly to climate change. MHW/thermal-exposure surfaces could inform siting and resilience monitoring of the network, but this is an inferred, not-yet-established use.
POL-022	30×30 Technical Working Group (marine)	1 LOW	Conceptual	Marine protected-area network design toward 30×30 could account for warming and MHW exposure to improve network resilience; a marginal, unproven use at present.
†	UK Blue Carbon Evidence Partnership (UKBCEP)	1 LOW	Conceptual	MHWs damage seagrass and saltmarsh and can alter coastal-carbon fluxes. Ocean-heat information could contextualise blue-carbon condition and permanence risk, but the link to the partnership's evidence work is speculative. Operational activity not held in the workbook (†).

Policy indicators

A scan of *indicators_climate_nature.xlsx* across its framework sheets for indicators whose measured quantity a UK marine heatwave product could directly inform — ocean heat / marine-heatwave intensity, and temperature-driven marine habitat, plankton and fish-stock condition — complemented by a focused web scan for authoritative indicators absent from the workbook. Pure GIS-area or fishing-pressure indicators (e.g. protected-area extent at sea, proportion of stocks harvested sustainably) were excluded because their measured quantity takes no ocean-temperature input. Items marked † are not held in the indicators workbook.

Indicator (ref)	Indicator name	Source framework	Current policy application(s)
IND-B-162	Ocean heat content and marine heatwaves in European seas	EEA Biodiversity / climate indicators	EU Climate Adaptation Strategy / EEA state-of-environment reporting; UK analogue via MCCIP. <i>The product's measured quantity is this indicator.</i>
IND-C-103	Marine and coastal habitat extent and condition (UK)	CCC Adaptation Monitoring Framework 2023–25	NAP3 / CCRA (Climate Change Act 2008, s.58)
IND-C-115	Fish stock status and catches in UK waters	CCC Adaptation Monitoring Framework 2023–25	NAP3 / CCRA; Fisheries Act 2020
IND-E-019	Diverse seas: condition of pelagic habitats (plankton community state)	Environmental Indicator Framework (EIF)	EIP 2025 (Environment Act 2021); UK Marine Strategy GES
IND-J-035	Pressure from climate change on biodiversity (species range exposure)	JNCC UK Biodiversity Indicators 2025	NBSAP 2025 ; Environment Act 2021 targets
†	MCCIP sea-temperature evidence update — UK & Ireland sea	MCCIP rolling evidence updates (web scan);	UK Marine Strategy Part One 2025 GES assessment; MCCIP

Indicator (ref)	Indicator name	Source framework	Current policy application(s)
	temperature (rolling evidence update; Annual Summary 2025). Met Office-supplied; overlaps EEA IND-B-162.	not in workbook)	report cards

Methods

1. **Capability scoped from external evidence**, not from MO outputs: the supplied bibliography is a marine-heatwave topic reading list, used to establish which MHW applications are demonstrated elsewhere (fisheries distribution, habitat/seagrass damage, blue-carbon and compound/sub-surface events) and how mature each is for UK policy use.
2. **MO/partner capability characterised** by inspecting MO, Cefas, Copernicus and EEA pages (accessed July 2026): OSTIA SST analysis, the FOAM / Copernicus Marine North-West Shelf system (3D temperature), the ESA SST_cci record, MO regional coupled prediction of MHW impacts on UK weather (2024), and the Cefas–MO *40 priority questions* MHW horizon-scan (npj Ocean Sustainability 2025).
3. **Policy candidates** drawn from a full sweep of the governance workbook's Policies & Activities sheet (uk_climate_nature_governance v18, 64 records; POL IDs stable across versions): screened across the Marine & Fisheries, Nature, Climate, Water and Land sectors for a plausible ocean-temperature dependency. Supplemented with operational marine activities absent from the workbook — the Fisheries Act 2020 climate objective / Joint Fisheries Statement & FMPs, aquaculture food-safety (HAB/*Vibrio*) monitoring, and the UK Blue Carbon Evidence Partnership — marked †. Purely terrestrial and non-marine records were excluded as out of scope.
4. **Scoring** used two independent axes. *Relevance* (materiality of a UK MHW product to the policy): central input = 3; supporting evidence layer = 2; marginal/indirect = 1. *Maturity*: Demonstrated where the application is proven elsewhere or already used and transferable; Emerging where partial evidence needs UK development; Conceptual where the use is plausible but unproven. Kept separate so a high-value-but-immature application (e.g. Fisheries Act/FMPs) is not mistaken for a low-relevance one.
5. **Indicator scan** of indicators_climate_nature.xlsx across its framework sheets, keyword-filtered for marine, sea-surface/ocean temperature, heatwave and fish/plankton, then narrowed to indicators whose measured quantity an MHW product could directly inform (ocean heat/MHW intensity, temperature-driven habitat/plankton/fish-stock condition); protected-area-extent and fishing-pressure indicators were excluded because their measured quantity takes no ocean-temperature input. A focused web scan for authoritative indicators absent from the workbook found the MCCIP UK & Ireland sea-temperature evidence update (overlaps EEA IND-B-162), marked †.

Main gaps and caveats

Three structural caveats. First, the bibliography is an external MHW reading list, not an MO publication list, so MO capability was scoped from MO/Cefas/Copernicus/EEA sources rather than the file. Second, the underpinning data (OSTIA, FOAM/CMEMS North-West Shelf, ESA SST_cci) already exist and MHW *monitoring* is Demonstrated; what is prospective is an **operational, categorised, forecast, policy-facing MHW product**, so most policy links here are Emerging or Conceptual rather than Current. Third, three of the highest-value marine activities are operational processes not held in the governance workbook — the Fisheries Act 2020 climate objective / Joint Fisheries Statement & FMPs, aquaculture food-safety (HAB/*Vibrio*) monitoring, and the UK Blue Carbon Evidence Partnership (marked †) — and are arguably worth adding as records. The HPMa, Marine Policy Statement/Plans, NCEA and blue-carbon links are credible but inferred, not yet evidenced in current policy text (Conceptual, lower confidence). Sub-surface and seafloor MHWs are a known monitoring gap even for the demonstrated uses.

References

Evidence base (external literature)

6. Holbrook, N.J., et al. (2019) A global assessment of marine heatwaves and their drivers. *Nature Communications* 10, 2624.
7. Oliver, E.C.J., et al. (2018) Longer and more frequent marine heatwaves over the past century. *Nature Communications* 9, 1324.
8. Cornes, R.C., et al. (2025) Climate change impacts on temperature around the UK and Ireland. *MCCIP Science Review* (rolling evidence update).
9. Plecha, S.M., Soares, P.M.M., Silva-Fernandes, S.M., et al. (2021) On the uncertainty of future projections of marine heatwave events in the North Atlantic Ocean. *Climate Dynamics* 56, 2027–2056.
10. Dzwonkowski, B., Coogan, J., Fournier, S., et al. (2020) Compounding impact of severe weather events fuels marine heatwave in the coastal ocean. *Nature Communications* 11, 4623.
11. Jung, E.M.U., Abdul Majeed, N.A.B., Booth, M.W., et al. (2023) Marine heatwave and reduced light scenarios cause species-specific metabolomic changes in seagrasses under ocean warming. *New Phytologist* 237, 1836–1851.
12. Ford, D.J., Nair, S.P., Oglethorpe, K., et al. (2025) Regionally different marine heatwave ocean carbon sink responses are consistent with carbonate understanding. *Environmental Research Communications*.

Met Office / partner capability and other sources

- Met Office, Operational Sea Surface Temperature and Ice Analysis (OSTIA) and FOAM / Copernicus Marine North-West Shelf ocean forecasting. metoffice.gov.uk/research/weather/ocean-forecasting
- ESA Climate Change Initiative — Sea Surface Temperature (SST_cci); Met Office science lead. climate.esa.int/.../sea-surface-temperature
- Pinnegar, J.K., Rowland, C., et al. (2025) 40 priority questions to advance understanding of the risks and opportunities of UK marine heatwaves. *npj Ocean Sustainability* 4, 20. nature.com/articles/s44183-025-00171-0
- Defra Marine Science blog (26 Sep 2024) Turning up the heat: advancing UK science to better predict and respond to marine heatwaves (Cefas / Met Office).
- Met Office (2025/2026) Waters surrounding UK gripped by prolonged marine heatwave; Long-lasting UK marine heatwave likely to reach extreme levels (media centre); National Oceanography Centre rolling updates.
- MCCIP (2025) Annual Summary 2025 and Sea Temperature rolling evidence update (Oct 2025). mccip.org.uk/recent-updates
- Defra (2025) Marine strategy part three: 2025 UK programme of measures; UK Marine Strategy Part One Update 2025 consultation (GOV.UK).
- Fisheries Act 2020 (c.22) and Joint Fisheries Statement 2022 (legislation.gov.uk / GOV.UK).
- EEA, European sea surface temperature and open-ocean marine-heatwave indicators; Copernicus C3S European State of the Climate 2024 — European Ocean.
- uk_climate_nature_governance.xlsx (v18), Policies & Activities sheet (links from the Link column).
- indicators_climate_nature.xlsx, framework indicator sheets (EIF, CCC, JNCC, EEA).